

ADC Architectures

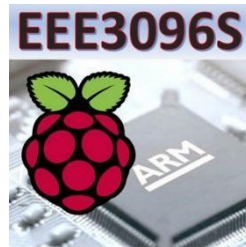
Sampling & Data Conversion · Module 4, Lecture 3 of 4

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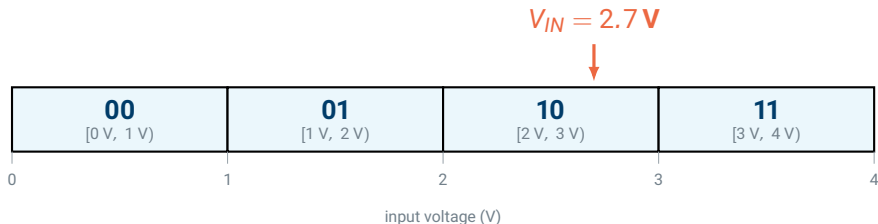


Which voltage interval contains the input?

$V_{REF} = 4\text{ V}$ $V_{IN} = 2.7\text{ V}$ held constant while this sample is converted $N = 2$, so

$$\Delta = V_{REF}/2^N = 1\text{ V}$$

Throughout: code = $\lfloor V_{IN}/\Delta \rfloor$, thresholds at whole multiples of Δ . Not the half-LSB-offset convention used in the ADC error diagrams.



What comparisons would identify the interval containing 2.7 V?

Different ADC architectures solve the same voltage-classification problem.

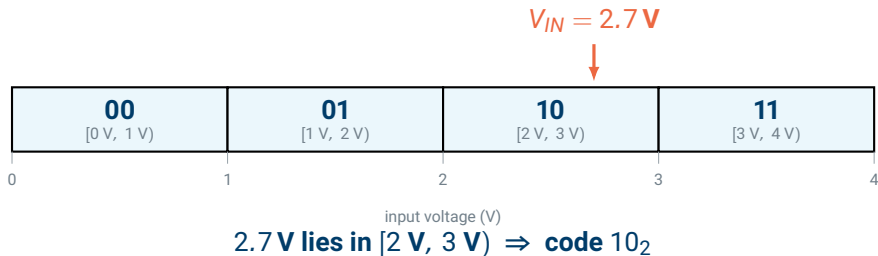


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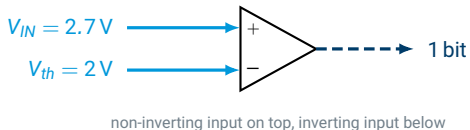


What comparisons would identify the interval containing 2.7 V?

Different ADC architectures solve the same voltage-classification problem.



A comparator answers one threshold question



Decision rule

$V_{IN} \geq V_{th} \Rightarrow$ output 1

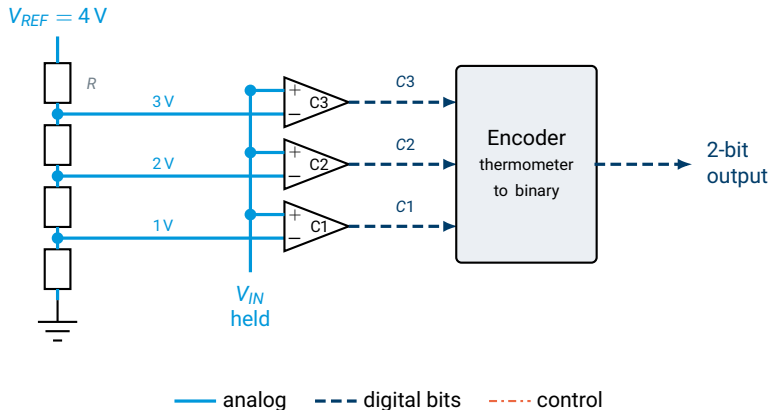
$V_{IN} < V_{th} \Rightarrow$ output 0

Here $2.7 \geq 2$, so this comparator outputs
1.

The architecture determines how comparator decisions become a multibit code.

One comparator reports only which side of one threshold the input is on.

Flash compares against every threshold in parallel



Dot = connection; a plain crossing is not a connection. Pattern order is written $C_1C_2C_3$, ascending threshold.

All threshold decisions are made in parallel. Comparator and encoder delays are small, not zero.

Board example: a two-bit flash conversion

$V_{REF} = 4\text{ V}$ $V_{IN} = 2.7\text{ V}$ thresholds at 1 V, 2 V, 3 V

Comparator question	Output
$V_{IN} \geq 1\text{ V}?$	
$V_{IN} \geq 2\text{ V}?$	
$V_{IN} \geq 3\text{ V}?$	

$C_1C_2C_3$	→	binary
000		00
100		01
110		10
111		11

Three comparator outputs form the thermometer pattern; the encoder produces the two output bits.



Board example: a two-bit flash conversion

$V_{REF} = 4\text{ V}$ $V_{IN} = 2.7\text{ V}$ thresholds at 1 V, 2 V, 3 V

Comparator question	Output
$V_{IN} \geq 1\text{ V}?$	1
$V_{IN} \geq 2\text{ V}?$	1
$V_{IN} \geq 3\text{ V}?$	0

$C_1C_2C_3$	→	binary
000		00
100		01
110		10
111		11

$$C_1C_2C_3 = 110$$

two thresholds crossed $\Rightarrow$ decimal 2

Three comparator outputs form the thermometer pattern; the encoder produces the two output bits.



Board example: a two-bit flash conversion

$$V_{REF} = 4\text{ V}$$

$$V_{IN} = 2.7\text{ V}$$

thresholds at 1 V, 2 V, 3 V

Comparator question	Output
$V_{IN} \geq 1\text{ V}?$	1
$V_{IN} \geq 2\text{ V}?$	1
$V_{IN} \geq 3\text{ V}?$	0

$C_1C_2C_3$	→	binary
000		00
100		01
110		10
111		11

$$C_1C_2C_3 = 110$$

two thresholds crossed $\Rightarrow$ decimal 2

110 thermometer $\Rightarrow$ **10₂** binary output

Three comparator outputs form the thermometer pattern; the encoder produces the two output bits.



Flash speed comes with a hardware cost

$$\text{number of comparators} = 2^N - 1$$

Resolution	Comparators
2 bits	3
4 bits	15
8 bits	255

Practical consequences

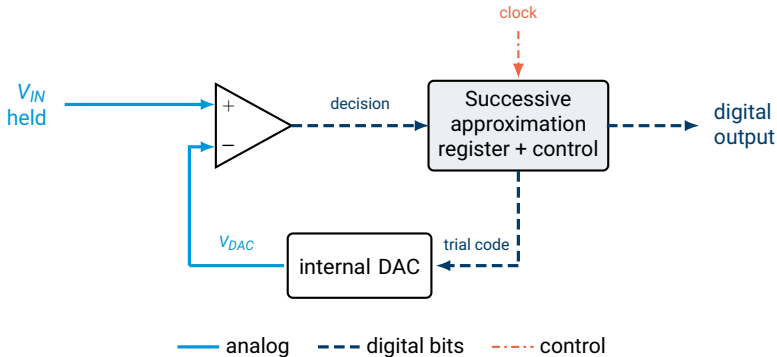
- Circuit area grows with the comparator bank.
- Every comparator loads the input node.
- Power scales with the number of active comparators.

Higher resolution rapidly increases the comparator count.

Could we reuse one comparator?



SAR tests one bit at a time



trial code $\rightarrow V_{DAC} \rightarrow$ comparator decision $\rightarrow$ SAR update

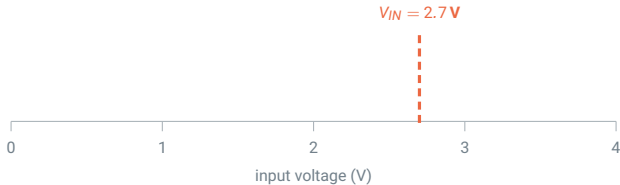
MSB first, then the remaining bits, down to the LSB. This is a binary search, not a count through every code.

SAR repeatedly narrows the input interval. The internal DAC converts a trial code into a comparison voltage.

Board example: a two-bit SAR conversion

$$V_{REF} = 4\text{ V} \quad V_{IN} = 2.7\text{ V} \quad V_{DAC} = \text{code} \times V_{REF}/2^N$$

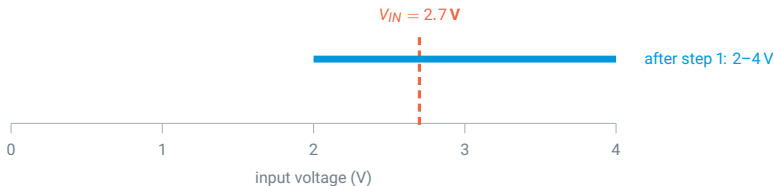
Step	Trial code	V_{DAC}	Decision
1 (MSB)			
2 (LSB)			



Board example: a two-bit SAR conversion

$$V_{REF} = 4\text{ V} \quad V_{IN} = 2.7\text{ V} \quad V_{DAC} = \text{code} \times V_{REF}/2^N$$

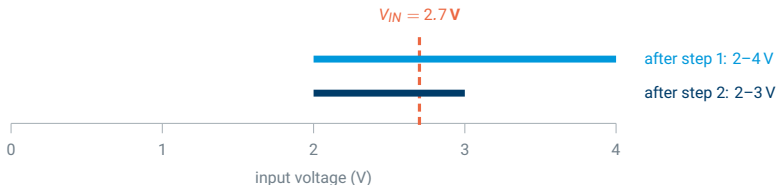
Step	Trial code	V_{DAC}	Decision
1 (MSB)	10_2	2 V	$2.7 \geq 2$, retain the MSB
2 (LSB)			



Board example: a two-bit SAR conversion

$$V_{REF} = 4\text{ V} \quad V_{IN} = 2.7\text{ V} \quad V_{DAC} = \text{code} \times V_{REF}/2^N$$

Step	Trial code	V_{DAC}	Decision
1 (MSB)	10_2	2 V	$2.7 \geq 2$, retain the MSB
2 (LSB)	11_2	3 V	$2.7 < 3$, clear the LSB



Final output 10_2 : the same result as flash, reached through successive decisions.



Flash and SAR organise comparisons differently

Flash

- Many comparators.
- Decisions taken in parallel.
- Tends towards high speed.
- Comparator count grows rapidly with N .

SAR

- One comparator in the basic architecture.
- N bit decisions for N bits, taken successively.
- Requires a trial DAC and control logic.
- Uses far fewer comparators.

For this two-bit example, both return 10_2 .

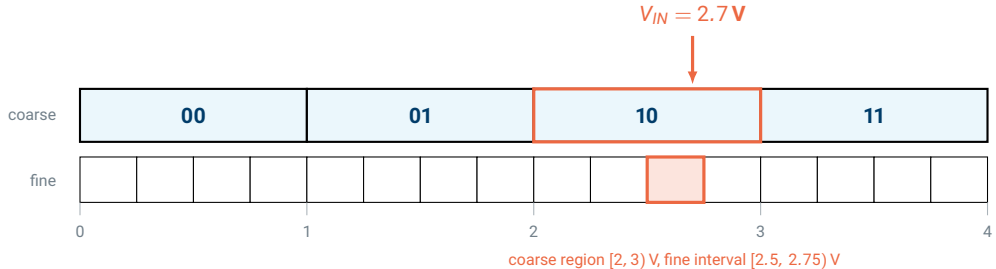
These are architectural tendencies. A given commercial part may not follow them.



Can we determine several bits at a time without a full-size flash bank?

Two-step conversion separates coarse and fine decisions

Four-bit resolution: 16 intervals across 0 to 4 V, each $\Delta = 0.25$ V wide.

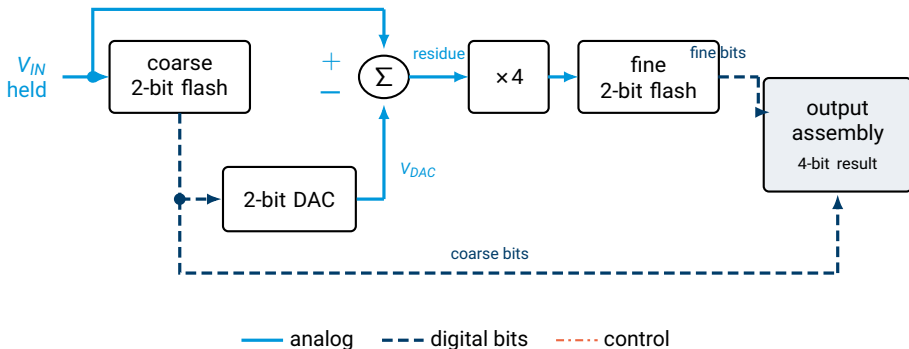


Residue: the part of the input not yet represented by the coarse result.

The first stage identifies a region; the second refines the position within it.



The second stage converts the residue



$$V_{\text{residue}} = V_{\text{IN}} - V_{\text{DAC}} \quad V_{\text{next}} = 4 V_{\text{residue}}$$

The coarse stage leaves a 0 to 1 V residue; a gain of four expands it to the fine stage's 0 to 4 V range.

Board example: two two-bit stages make four bits

$$V_{IN} = 2.7\text{ V} \quad V_{REF} = 4\text{ V}$$

Step	Value
Coarse conversion	
Coarse DAC	
Residue	
Amplified residue	
Fine conversion	
Assembled output	

The DAC, subtractor and amplifier are additional requirements. Comparator count alone is not a complete area or power comparison.



Board example: two two-bit stages make four bits

$$V_{IN} = 2.7\text{ V} \quad V_{REF} = 4\text{ V}$$

Step	Value
Coarse conversion	$2.7\text{ V} \rightarrow 10_2$
Coarse DAC	$10_2 \rightarrow 2\text{ V}$
Residue	$2.7 - 2 = 0.7\text{ V}$
Amplified residue	
Fine conversion	
Assembled output	

The DAC, subtractor and amplifier are additional requirements. Comparator count alone is not a complete area or power comparison.



Board example: two two-bit stages make four bits

$$V_{IN} = 2.7\text{ V} \quad V_{REF} = 4\text{ V}$$

Step	Value
Coarse conversion	$2.7\text{ V} \rightarrow 10_2$
Coarse DAC	$10_2 \rightarrow 2\text{ V}$
Residue	$2.7 - 2 = 0.7\text{ V}$
Amplified residue	$0.7 \times 4 = 2.8\text{ V}$
Fine conversion	$2.8\text{ V} \rightarrow 10_2$
Assembled output	

The DAC, subtractor and amplifier are additional requirements. Comparator count alone is not a complete area or power comparison.



Board example: two two-bit stages make four bits

$$V_{IN} = 2.7\text{ V} \quad V_{REF} = 4\text{ V}$$

Step	Value
Coarse conversion	$2.7\text{ V} \rightarrow 10_2$
Coarse DAC	$10_2 \rightarrow 2\text{ V}$
Residue	$2.7 - 2 = 0.7\text{ V}$
Amplified residue	$0.7 \times 4 = 2.8\text{ V}$
Fine conversion	$2.8\text{ V} \rightarrow 10_2$
Assembled output	$10 10 \rightarrow \mathbf{1010_2}$

Check

$1010_2 = 10$, and $10 \times 0.25\text{ V} = 2.5\text{ V}$.

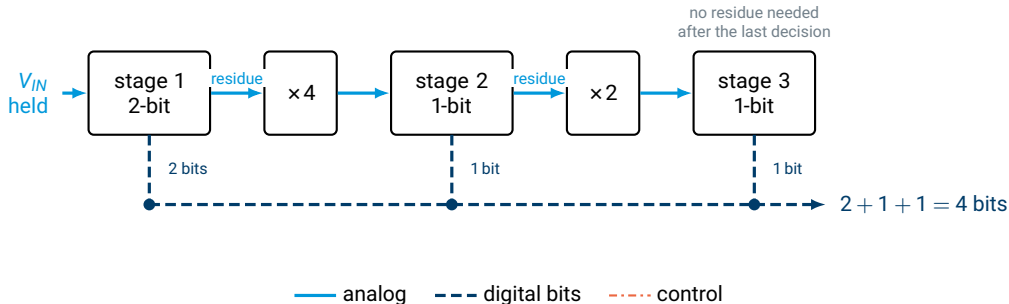
2.7 V lies in $[2.5, 2.75)\text{ V}$. ✓

Two two-bit stages: $3 + 3 = 6$ comparators.
One four-bit flash: 15 comparators.

The DAC, subtractor and amplifier are additional requirements. Comparator count alone is not a complete area or power comparison.



Multistage conversion repeats the refinement



Each one-bit stage compares against $2V$; where a one-bit DAC is required it reconstructs either $0V$ or $2V$.

Later stages refine the residue left by earlier stages. No stage measures the original V_{IN} again.



Board example: four bits from three stages

$$V_{IN} = 2.7\text{ V} \quad \text{one-bit threshold} = 2\text{ V}$$

Stage	Input	Result	DAC	Next-stage input
1	2.7 V			
2				
3				

bits so far:



Board example: four bits from three stages

$$V_{IN} = 2.7\text{ V} \quad \text{one-bit threshold} = 2\text{ V}$$

Stage	Input	Result	DAC	Next-stage input
1	2.7 V	10	2 V	$(2.7 - 2) \times 4 = 2.8\text{ V}$
2	2.8 V			
3				

bits so far: **10**



Board example: four bits from three stages

$$V_{IN} = 2.7\text{ V} \quad \text{one-bit threshold} = 2\text{ V}$$

Stage	Input	Result	DAC	Next-stage input
1	2.7 V	10	2 V	$(2.7 - 2) \times 4 = 2.8\text{ V}$
2	2.8 V	1	2 V	$(2.8 - 2) \times 2 = 1.6\text{ V}$
3	1.6 V			

bits so far: **10 1**



Board example: four bits from three stages

$$V_{IN} = 2.7\text{ V} \quad \text{one-bit threshold} = 2\text{ V}$$

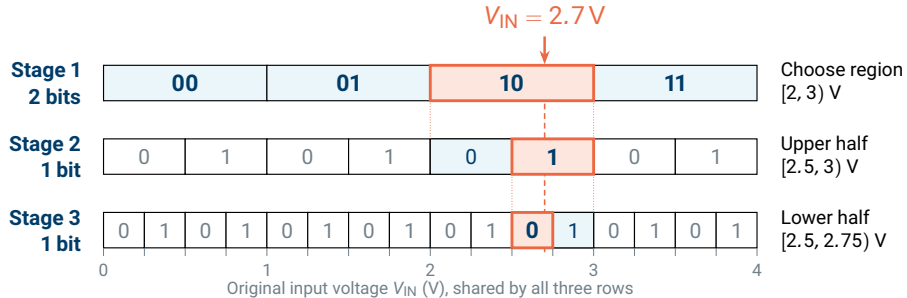
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2	2.8 V	1	2 V	$(2.8 - 2) \times 2 = 1.6\text{ V}$
3	1.6 V	0	–	–

bits so far: **10 1 0** $\Rightarrow$ 10|1|0 = **1010₂**

The same four-bit result as the two-step example, reached with one 2-bit and two 1-bit decisions.



Three-stage conversion narrows the interval



$$\underbrace{10}_{\text{stage 1}} \mid \underbrace{1}_{\text{stage 2}} \mid \underbrace{0}_{\text{stage 3}} = 1010_2$$

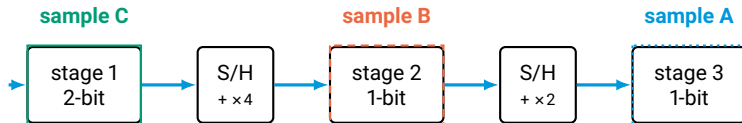
For stages 2 and 3: 0 selects the lower half; 1 selects the upper half.
The axis shows input ranges, not the amplified residue voltages inside the ADC.

Each stage halves the interval that survived the stage before it. Only the selected region is ever refined.



Pipelining overlaps different samples

After stage 1 passes sample A's residue onward, can it start sample B?



one instant in time: three different samples are in flight at once

Multistage

Divides *one* conversion into stages.

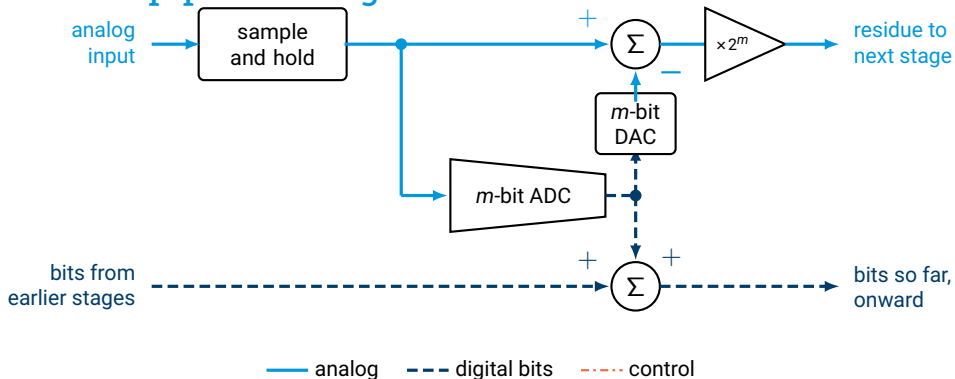
Pipelined

Overlaps work on *several* samples.

Because every stage holds its own input, p conversions can be underway at once.



Inside one pipeline stage



Hold the input, resolve m bits, reconstruct them, subtract, amplify what is left, and pass on *both* the residue and the bits.

Every stage carries its own sample-and-hold. That is what frees a stage to start the next sample.



Staging trades cycles for comparators

p stages of m bits give $n = p \times m$ bits, using $p(2^m - 1)$ comparators.

Resolution	One flash bank	As a pipeline	Comparators
8 bits	255 comparators	2 stages $\times$ 4 bits	30
16 bits	65 535 comparators	4 stages $\times$ 4 bits	60

Cycles to convert one sample: flash **1** · pipeline **p** · SAR **n**

A 16-bit pipeline needs roughly a thousandth of a 16-bit flash bank's comparators, paid for in latency rather than throughput.

Real parts add a few bits per stage for error correction; these are the ideal minimum.



Latency and throughput measure different things

Simplified model: one stage takes one clock interval, and a completed output appears at the end of the interval shown.

Clock	Stage 1	Stage 2	Stage 3	Completed output
1	A	–	–	–
2	B	A	–	–
3	C	B	A	A
4	D	C	B	B
5	E	D	C	C

Latency

Sample A enters at clock 1 and its result appears at clock 3: **three clock intervals**, one per stage.

Throughput

Once the pipeline is full, **one result per clock**: the same throughput as a flash converter.

Delay the earlier-stage bits so that the assembled bits all belong to the same sample.

A pipeline wants a fixed clock period, so non-periodic inputs suit SAR better. Real timing is implementation dependent.



Compare the conversion strategies

Architecture	How it decides	Main tradeoff
Flash	Parallel threshold comparisons	Many comparators
SAR	Successive bit decisions	Sequential decision time
Two-step	Coarse conversion plus residue refinement	Additional analog processing
Multistage	Repeated residue refinement	Interstage accuracy and coordination
Pipeline	Concurrent processing of different samples	Output latency and sample alignment

- a. Why does a two-bit flash ADC need three comparators?
- b. Why does a SAR ADC contain a DAC?
- c. How can a pipeline deliver one result per clock when each sample takes several clocks?



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- a. Why does a two-bit flash ADC need three comparators?
Four intervals are separated by three internal boundaries: 1 V, 2 V and 3 V.
- b. Why does a SAR ADC contain a DAC?
To turn each trial code into a voltage the single comparator can compare against.
- c. How can a pipeline deliver one result per clock when each sample takes several clocks?
Stages work on different samples at once: latency stays, throughput rises.



Questions?

End of Lecture 4.3 · ADC Architectures

Next: L8, DACs and Analogue Signal Generation

